

B2-PROPA-HLAYER claim-level synthesis: Proposition A and the discharge of H-LAYER

TECT verification-first repository · claim B2-PROPA-HLAYER

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1. Purpose and scope

Claim-level synthesis for B2-PROPA-HLAYER. It adds no mathematics; every statement carries the unit's registered tier. **Scope:** the composition of B2's units into the canonical claim state. **Not claimed:** anything beyond the enacted scope; in particular Prop-A's own bundle scope is $\mathcal{A}_{\text{adm}}$ ($T' \leq 13$) and the C_{full} statement runs through the joint B1/B2 head package, not through Prop-A alone.

2. The claim and its canonical state

Claim (canonical). The isotropic Gaussian–Hartree layer is the strict comparison infimum: the H-LAYER hypothesis class is DISCHARGED – every analytic H-LAYER residual is closed class-wide on the primary admissible class, and the layer comparison enters the Reading-H selection as a theorem, not an assumption. Tier: T7-SCOPE $_{C_{\text{full}}}$ (jointly with B1, enacted 2026-06-10); hypothesis: A1-KERNEL-CONV; open gates: none.

3. Sub-proof unit map (per-unit tier and role)

Unit	Tier	What it contributes	Publication state
Prop-A/	T6	(D) block-diagonal isotropy (entropy floor $\frac{1}{2}r_R^2 > 0$, breathing $\frac{3}{2}u_{\text{eff}} > 0$) + (O) class-wide closure on $\mathcal{A}_{\text{adm}}$ ($R_{\text{lead}}(13) = 0.650 < 1$, joint = 1.097 > 1, $T'_{\text{crit}} = 60.4$); T-016..T-024 chain	PUBLISHED – Prop-A-T6-260611 (9/9 scripts, 44/44)
G-A0-DUI/	T6/T7-comp.	closed lemma: $M'(m) < 0$ by differentiation under the integral (dominated convergence) – supports $A = 0$ uniqueness of the reference	AUXILIARY (operator-accepted v1.1; candidate fold into Prop-A/A1)
H-A0-removal/	T6/T7-comp.	sign-decomposition theorem $F'_0 = \frac{1}{2}M'g$ discharging H-A0's U, Z to H-ANCHOR within the certified window (zero-at-gap anchor $g(m_{\text{gap}}) = 0$)	AUXILIARY (operator-accepted v1.1; candidate fold into Prop-A/A1)

External: the joint B1/B2 head package Reading-H-cFull-T7-260611 carries the C_{full} extension; Prop-A is its (D)/(O) primary-class sub-structure ($\mathcal{A}_{\text{adm}} \subset \mathcal{C}_{\text{full}}$).

4. Composition and what remains

Prop-A (T6, PUBLISHED) closes every analytic H-LAYER residual class-wide on $\mathcal{A}_{\text{adm}}$ at the operating point via the three sectors (D)/(O)/(S); the two $A = 0$ lemmas secure the reference-state uniqueness feeding the A1 convention; the C_{full} statement is enacted at the head. Nothing in B2's gate list remains open. Remaining (non-blocking, registered): the operator fold-in decision for G-A0-DUI/H-A0-removal

into Prop-A or the A1 record (cosmetic restructuring); the A1-KERNEL-CONV migration itself (claim-A1-side, per main-proof-line.md).

5. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3)

- (α) “B2 is T7-SCOPE but its only published in-claim bundle is T6 – mismatch?” DISMISSED with explanation: the T7-SCOPE state is the JOINT B1/B2 head enactment (Reading-H package); Prop-A’s T6 is its primary-class sub-structure and was deliberately NOT re-tiered upward (operator ruling 2026-06-11: ‘T6 NOT T7, does not independently enact full C_full’). The claim-level tier and the unit-level tier answer different questions; both are quoted.
- (β) “The $A = 0$ lemmas might be load-bearing yet only operator-accepted drafts.” VALID with mitigation: they support the A1 REFERENCE definition (input layer), not the comparison chain; A1 itself is the registered hypothesis of both B1 and B2, so their weight is carried by the hypothesis declaration, not silently.
- (γ) “T-019 history (hdiag_gershgorin_rowsum exchange-sign framing) was superseded – does the synthesis cite stale framing?” DISMISSED: the superseded framing was replaced by the res5_019 local-functional reframing ($f''(M) = \frac{3}{2}u_{\text{eff}} > 0$) and the bundle carries the corrected note (Prop-A v1.3, T-019 supersession note retained as history).

Quantitative sanity check (mandatory). The quoted Prop-A load-bearing values ($\Phi''_{\text{diag}} = 4.0275 > 0$, $\frac{1}{2}r_R^2 = 4.6368 \times 10^{-2} > 0$, $R_{\text{lead}}(13) = 0.650 < 1$, $\text{joint}(13) = 1.097 > 1$, $T'_{\text{crit}} = 60.4$) are the operator-confirmed bundle values (9/9, 44/44); margins 0.350 and 0.097 are positive; the $\mathcal{A}_{\text{adm}} \subset \mathcal{C}_{\text{full}}$ inclusion is definitional ($T' \leq 13$ vs the class cap).

6. Result footer

Result ID:	B2-PROPA-HLAYER / claim-level SYNTHESIS (T-013)
Precise statement:	B2’s canonical state (T7-SCOPE_{C_full} given A1; H-LAYER discharged; gates none) is carried by the joint B1/B2 head package plus the PUBLISHED Prop-A-T6-260611 primary-class sub-structure; G-A0-DUI and H-A0-removal are auxiliary A=0-reference lemmas feeding A1.
Scope:	synthesis layer; no new math, no tier change, no gate flip.
Evidence grade:	SYNTHESIS over PUBLISHED + operator-accepted units.
Reproduction command:	run the Prop-A bundle entry manifest (9 scripts); head package per its bundle.
Falsifier:	any falsifier of the cited footers; or a unit-tier mismatch against INDEX.md.
Tier before / after:	no change (B2 stays T7-SCOPE_{C_full}; Prop-A unit stays T6).
No-overclaim statement:	Prop-A alone does not enact C_full; the claim-level tier is the joint head enactment.
Next required action:	operator: optional fold-in decision for the two A=0 lemmas (cosmetic restructuring).